

Functions

In Python, a function is a block of organized, reusable code that performs a specific task or set of tasks. Functions allow you to break down your code into smaller, modular pieces, making it more readable, maintainable, and reusable.

Defining a Function:

To define a function, you use the

`def` keyword, followed by the function name and a pair of parentheses. If the function takes parameters, you list them within the parentheses. The function code is indented below the definition.

```
def greet():  
    print("Hello, welcome to pythonlife")
```

- **Function Call:**

To execute a function and perform the tasks defined within it, you call the function by using its name followed by parentheses. If the function expects parameters, you provide them within the parentheses.

```
greet()
```

- **Return Statement:**

Definition: The `return` statement is used to exit a function and return a value to the caller.

```
def add(x, y):  
    return x + y  
  
result = add(3, 5) # result is now 8  
print("The result is:", result) # Output: The result is: 8
```

- **Parameters and Arguments:**

Parameters are variables that are used in a function definition, while

arguments are values passed to the function during the function call. Parameters receive values from arguments.

```
def multiply(x, y):  
    return x * y  
multiply(3, 4) # Here, 3 and 4 are arguments
```

- **Default Parameters:**

You can provide default values for parameters, which allows the function to be called with fewer arguments. If a value is not provided for a default parameter, the default value is used.

```
def power(base, exponent=2):  
    return base ** exponent  
  
power(3)          # Uses default exponent (2)  
power(3, 4)       # Uses provided exponent (4)
```

Module

In programming, a module is a file containing Python definitions and statements. These files typically have a `.py` extension and are used to organize code into reusable units.

Here are a few examples of Python modules along with a brief explanation of each:

math:

The `math` module provides mathematical functions such as trigonometric, logarithmic, exponential functions, and more.

```
import math  
  
print(math.sqrt(16))    # Square root  
print(math.pi)         # Output: 3.141592653589793  
print(math.pow(2, 3))   # Output: 8.0
```

random:

The `random` module is used for generating pseudo-random numbers.

```
import random

print(random.randint(1, 10)) # Random integer between 1 and 10
print(random.choice(['apple', 'banana', 'orange'])) # Random choice
```

datetime:

The `datetime` module provides classes for working with dates and times.

```
from datetime import datetime

now = datetime.now()
print(now) # Current date and time
```

os:

The `os` module provides a way to interact with the operating system, such as reading or changing the current working directory.

```
import os

print(os.getcwd()) # Get current working directory
```

sys:

```
# import sys
# sys.path.insert(0, r"path")
```